

Event Budget Planning and Expense Approval System

Team C13

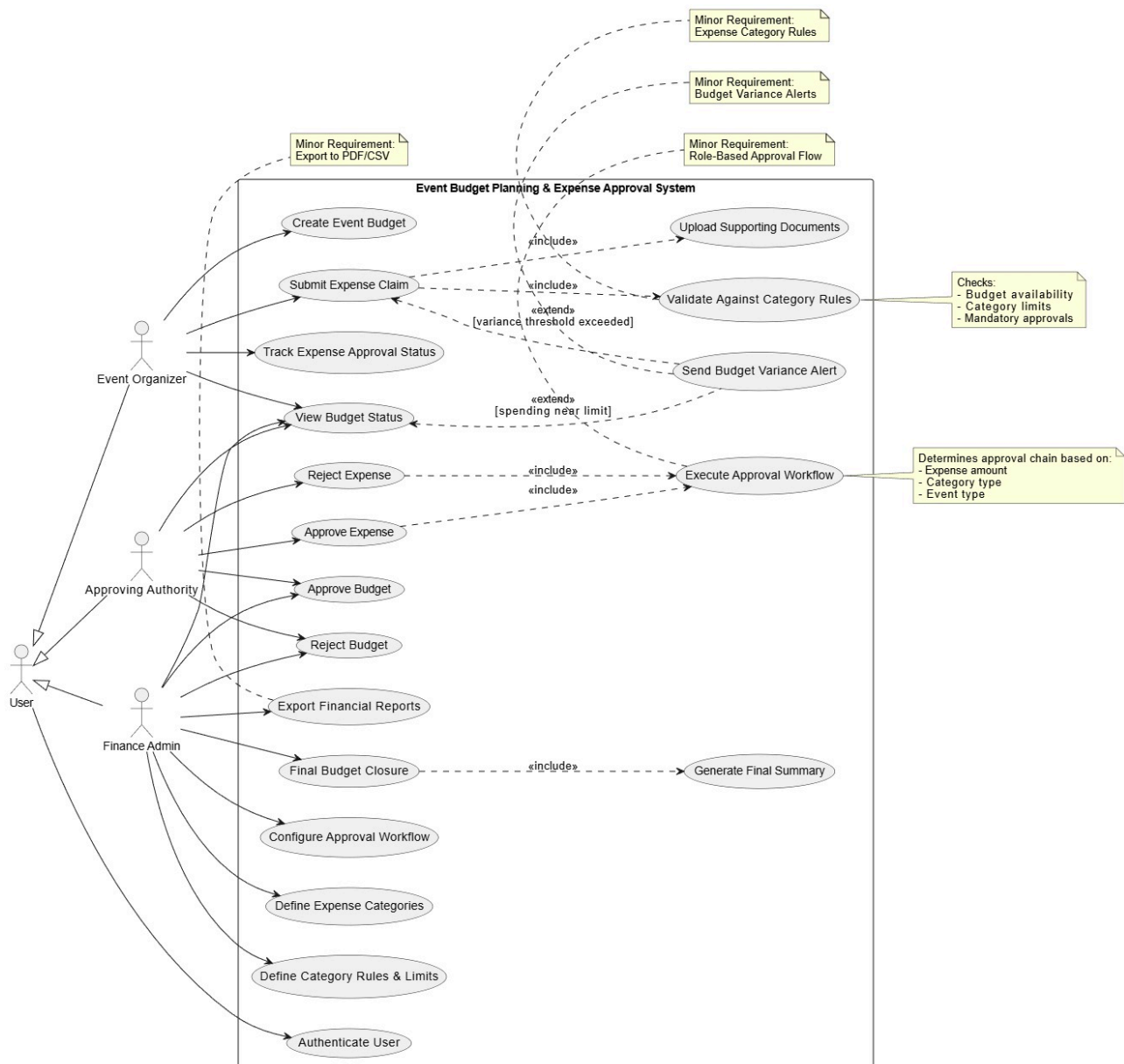
Chiyedu Vishnu : PES1UG23CS169

Kaustubh C : PES1UG23CS154

Dareddy Devesh Reddy : PES1UG23CS172

BB Varun Kumar : PES1UG23CS131

USE CASE Diagram



USE CASE SPECIFICATIONS

USE CASE 1: CREATE EVENT BUDGET

Name

Create Event Budget

Summary

Event Organizer creates a new event budget by defining budget categories, allocating amounts to each category, and submitting the budget for approval by Approving Authorities.

Actors

Primary Actor: Event Organizer **Secondary Actors:** Approving Authority, System

Pre-conditions

- Event Organizer is authenticated and has permission to create budgets
- Expense categories are already defined in the system by Finance Admin

Description

Event Organizer accesses the budget creation form, enters event details (name, date, description, total budget), and allocates amounts to predefined expense categories (Venue, Food, Marketing, Equipment, etc.). System validates that category allocations sum equals total budget, assigns a unique Budget ID, sets status to "Pending Approval", and notifies the designated Approving Authority for review.

Exceptions

- **E1: System Validation Failure** - System displays field-specific errors and prevents submission until corrected
- **E2: Database Connection Error** - System retains data in session and allows retry without data loss
- **E3: Notification Service Failure** - Budget is saved successfully but notification is queued for retry

Alternate Flows

- **A1: Total Allocation Mismatch** - System displays error showing difference and prevents submission until allocations equal total budget
- **A2: Save as Draft** - Budget is saved with status "Draft" without triggering approval workflow or notifications
- **A3: Budget Exceeds Department Limit** - System flags budget for higher-level approval and requires additional justification
- **A4: Cancel Budget Creation** - System prompts for confirmation and discards all entered data if confirmed

Post-conditions

Success: Budget record created with unique ID, status set to "Pending Approval", visible to organizer and approvers, notification sent to Approving Authority, and audit trail entry logged.

Failure: No budget created, no notifications sent, system returns to stable state with error logged.

USE CASE 2: SUBMIT EXPENSE CLAIM

Name

Submit Expense Claim

Summary

Event Organizer submits an expense claim against an approved budget by providing expense details, uploading supporting documents, and routing the claim through the configured approval workflow.

Actors

Primary Actor: Event Organizer **Secondary Actors:** Approving Authority, Finance Admin, System

Pre-conditions

- Event Organizer is authenticated and event budget is in "Approved" status
- Budget is not closed and selected category has available balance

Description

Event Organizer selects an approved budget, chooses expense category, enters details (date, vendor, amount, description), and uploads mandatory supporting documents (invoice/receipt). System validates expense against category rules and remaining balance, marks amount as

"committed", determines appropriate approval workflow based on amount and category, routes to designated approver, and sends notification.

Exceptions

- **E1: Insufficient Budget Balance** - System blocks submission and suggests reducing amount or requesting reallocation
- **E2: Budget Closed During Submission** - System displays error and prevents submission completely
- **E3: Document Upload Service Unavailable** - System offers to save as draft without documents or wait and retry
- **E4: Database Connection Error** - System retains data in session and uploaded documents for retry

Alternate Flows

- **A1: Category Rule Violation** - System displays violated rule and prevents submission until expense complies
- **A2: Mandatory Document Missing** - System blocks submission with highlighted upload section until invoice/receipt uploaded
- **A3: Save Expense as Draft** - Expense saved without committing budget or triggering workflow
- **A4: Budget Variance Alert Triggered** - System sends alerts to organizer and Finance Admin when category reaches 80% utilization
- **A5: Split Expense Across Categories** - System validates each split separately and handles as single expense with multiple line items

Post-conditions

Success: Expense created with unique ID and "Pending Approval" status, amount marked as committed, documents uploaded, routed to approver with notification sent, and audit trail logged.

Failure: No expense created, budget unchanged, no documents uploaded, and error logged.

USE CASE 3: APPROVE OR REJECT EXPENSE

Name

Approve or Reject Expense

Summary

Approving Authority reviews submitted expense claims and either approves them (converting committed amounts to actual expenditure) or rejects them (releasing committed amounts back to available balance).

Actors

Primary Actor: Approving Authority **Secondary Actors:** Event Organizer, Next Approver (if multi-level), System

Pre-conditions

- Approving Authority is authenticated and expense is in "Pending Approval" status
- Expense is assigned to approver's queue and approver has sufficient authorization level
- Budget is not closed

Description

Approving Authority views pending approvals list, selects an expense to review, examines details including event context, budget balances, supporting documents, and organizer history. System verifies authorization level, approver makes decision to approve or reject, enters optional/mandatory comments, and system executes workflow (either final approval updating balances, or routing to next level, or rejection releasing committed amount), sends notifications to relevant stakeholders.

Exceptions

- **E1: Insufficient Authorization** - System displays error and disables approve/reject buttons, enabling only delegation option
- **E2: Budget Closed During Review** - System auto-rejects expense and notifies organizer
- **E3: Concurrent Approval Conflict** - System detects simultaneous modification and prompts approver to refresh
- **E4: Database Connection Error** - System retains decision in session and offers retry options

Alternate Flows

- **A1: Multi-Level Approval Required** - System routes to next approver instead of final approval and notifies next level
- **A2: Request Additional Information** - System changes status to "Information Requested", removes from queue, and notifies organizer
- **A3: Delegate Approval** - System reassigns expense to selected approver with logged reason
- **A4: Bulk Approval** - System processes multiple selected expenses simultaneously with single comment applied to all
- **A5: Partial Approval (Split Expense)** - System approves some portions and rejects others separately

Post-conditions

Success (Approval): Expense status "Approved", committed amount converted to approved expenditure, notification sent to organizer (and next approver if multi-level), audit trail logged.

Success (Rejection): Expense status "Rejected", committed amount released to available balance, notification with reason sent to organizer, audit trail logged. **Failure:** Expense status unchanged, budget unchanged, error logged, approver can retry.

USE CASE 4: FINAL BUDGET CLOSURE

Name

Final Budget Closure

Summary

Finance Admin closes the event budget after event completion, preventing further modifications, generating comprehensive final reports with variance analysis, and marking unspent amounts for reallocation.

Actors

Primary Actor: Finance Admin **Secondary Actors:** Event Organizer, Finance Committee, System

Pre-conditions

- Finance Admin is authenticated and event has concluded (event date passed)
- All expense claims are in final status (no pending approvals)
- Budget is in "Approved" status

Description

Finance Admin views eligible budgets for closure, selects a budget, reviews pre-closure validation (no pending expenses, event concluded), examines complete expense summary and variance analysis. System generates final budget report including allocation vs actual tables, variance explanations, approval timeline, and complete expense lists, updates status to "Closed", marks budget as read-only, flags unspent amounts for reallocation, and sends closure notifications with final report to organizer and finance committee.

Exceptions

- **E1: Pending Approvals Exist** - System displays error listing pending expenses and offers options to wait, notify approvers, or force close with auto-rejection
- **E2: Database Connection Error** - System retains closure data and offers retry without re-entering information
- **E3: Report Generation Failure** - System offers to close without report and queue generation for later
- **E4: Notification Service Failure** - Budget closed successfully but notifications queued for retry

Alternate Flows

- **A1: Event Not Yet Concluded** - System warns and requires additional justification for early closure
- **A2: High Variance Detected** - System requires mandatory explanation for categories exceeding $\pm 20\%$ variance
- **A3: Partial Budget Utilization** - System alerts for $>30\%$ unspent amount and prompts verification before marking for reallocation
- **A4: Force Close with Pending Expenses** - System auto-rejects all pending expenses with documented reason and notifies affected organizers
- **A5: Export Final Report** - System generates PDF (formatted with charts) or CSV (raw data) for download

Post-conditions

Success: Budget status "Closed", budget is read-only, final summary generated with variance analysis, unspent amounts marked for reallocation, closure notifications sent, and audit trail logged. **Failure:** Budget status unchanged, remains open for expense submissions, error logged, Finance Admin can retry closure.
